SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 1 – Case Study Analysis

Course Code:	CSA10 / CSA1010	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 1 – Case Study Analysis
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
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Date:	5.10.26	**Duration:**	60 minutes

Code of Conduct

I Bhavya S(192524376) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____

Annexure A – Case Study Analysis

Instructions to Students:

Each student will be allotted **one problem statement**. Analyze the assigned problem systematically by following the steps given below. Your report should demonstrate your understanding of software engineering concepts and your ability to design an appropriate software solution.

Based on your allotted problem statement, prepare a report by answering the following questions:

1. **Understand the Problem Statement**
   - Explain the problem in your own words.
   - Identify the objectives of the proposed system.
   - State the scope of the problem.
2. **Identify Key Stakeholders and Challenges**
   - Identify all the stakeholders involved in the system.
   - Explain the roles and responsibilities of each stakeholder.
   - Discuss the major challenges and issues faced by the stakeholders.
3. **Analyze the Current Approach**
   - Describe the existing system or current process.
   - Identify its strengths and limitations.
   - Analyze the problems or gaps that need to be addressed.
4. **Propose a Suitable Solution Using Software Engineering Principles**
   - Design a software-based solution for the given problem.
   - Describe the major functional modules of the proposed system.
   - Identify the functional and non-functional requirements.

- Explain how software engineering principles such as modularity, scalability, maintainability, reliability, usability, and security are incorporated into your solution.

5. **Justify the Chosen Methodology/Framework**
   - Select an appropriate Software Development Life Cycle (SDLC) model or software development framework (e.g., Agile, Scrum, Waterfall, Spiral, DevOps).
   - Justify why the selected methodology/framework is suitable for the proposed system.
   - Explain how it supports successful project development and maintenance.
6. **Discuss Expected Outcomes and Limitations**
   - Describe the expected benefits of the proposed solution.
   - Explain how the solution addresses the identified challenges.
   - Discuss potential risks, implementation challenges, and limitations.
   - Suggest possible future enhancements.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Problem Understanding (30%)	Clearly explains the problem, objectives, and scope with complete understanding.	Explains the problem with minor omissions.	Basic understanding with limited explanation.	Poor understanding; objectives and scope are unclear or missing.
Stakeholder & Challenge Analysis (25%)	Identifies all stakeholders, their roles, and challenges with clear analysis.	Identifies most stakeholders and challenges with adequate analysis.	Identifies only a few stakeholders or challenges.	Stakeholders and system analysis are incomplete or inaccurate.
Current System Analysis (20%)	Thoroughly analyzes the existing system, strengths, weaknesses, and identifies all gaps.	Good analysis with most strengths and weaknesses identified.	Limited analysis with partial identification of issues.	Proposed solution is unclear or inappropriate; requirements are largely missing.

Proposed Software Solution & Software Engineering Principles (15%)	Proposes an innovative, feasible solution applying appropriate software engineering principles (modularity, scalability, maintainability, security, etc.).	Suitable solution with good application of software engineering principles.	Basic solution with limited application of software engineering principles.	Methodology is inappropriate, poorly justified, or not discussed.
Methodology/ Framework Justification (10%)	Selects the most suitable SDLC/model/framew ork and provides strong technical justification.	Suitable methodology selected with adequate justification.	Methodology selected but justification is weak.	Outcomes, limitations, or future enhancements are missing; report lacks organization and clarity.

Criteria	**Mark**
Problem Understanding (30%)	/5
Stakeholder & Challenge Analysis (25%)	/5
Current System Analysis (20%)	/5
Proposed Software Solution & Software Engineering Principles (15%)	/5
Methodology/ Framework Justification (10%)	/5
TOTAL	/25

PROBLEM STATEMENT:

Personal Expense and Budget Management System: Develop an application that allows users to

record income and expenses, categorize spending, create budgets, receive alerts, and view financial reports.

CASE STUDY ANALYSIS

PERSONAL EXPENSE AND BUDGET MANAGEMENT SYSTEM

1. UNDERSTAND THE PROBLEM STATEMENT

1.1 Problem Description

Managing personal finances manually can be difficult because users may not accurately record their income and expenses, track spending categories, or monitor whether they are following their budgets. Expenses may be recorded in notebooks, spreadsheets, or different applications, making financial information difficult to organize and analyze.

The proposed Personal Expense and Budget Management System is a software application that enables users to record income and expenses, categorize spending, create budgets, receive alerts, and view financial reports. The system provides a centralized platform for managing personal financial information.

1.2 Objectives

The main objectives of the system are:

1. To allow users to record and manage their income.
2. To record daily expenses easily.
3. To categorize expenses such as food, transport, education, shopping, bills, and entertainment.
4. To allow users to create monthly or category-wise budgets.
5. To monitor actual spending against the planned budget.
6. To provide alerts when spending approaches or exceeds the budget limit.
7. To generate financial reports for better understanding of spending patterns.
8. To help users make informed financial decisions.
9. To provide secure and reliable access to personal financial information.

1.3 Scope of the System

The system covers:

- User registration and login.
- Income management.
- Expense management.
- Expense categorization.
- Budget creation and modification.
- Budget monitoring.
- Budget limit alerts.
- Financial reports.
- Spending summaries.

- Secure storage of user information.

The system can be developed as a web or mobile application and can be extended in the future with features such as financial predictions and automatic transaction import.

2. KEY STAKEHOLDERS AND CHALLENGES

2.1 Stakeholders

Stakeholder	Role and Responsibility
Users	Add income and expenses, create budgets, monitor spending and view reports.
System Administrator	Manage users, maintain the system and monitor system operations.
Software Developers	Design, develop, test and maintain the application.
Database Admin	Maintain the database, backup data and ensure data integrity.
Project Manager	Plans the project, manages resources and monitors project progress.
Testing Team	Tests functionality, security, usability and reliability of the application.

2.2 Major Challenges

Challenge 1 – Accurate Data Entry

Users may forget to record transactions or enter incorrect amounts.

Solution: Provide a simple and user-friendly transaction entry interface with validation.

Challenge 2 – Budget Overspending

Users may spend more than their planned budget without realizing it.

Solution: The system continuously compares expenses with budget limits and generates alerts. **Challenge 3 – Financial Data Security**

Personal financial information is sensitive and must be protected.

Solution: Implement authentication, authorization, secure password storage and protected database access.

Challenge 4 – Large Number of Transactions

Users may have hundreds or thousands of transactions over time.

Solution: Use an organized database with efficient searching, filtering and categorization. **Challenge 5 – Understanding Spending Patterns**

Raw transaction records may not clearly show where money is being spent.

Solution: Generate summaries and reports based on categories, dates and budgets.

Challenge 6 – User Engagement

Users may stop maintaining their financial records if the application is complicated. **Solution:** Provide a simple interface, reminders and meaningful alerts.

3. ANALYSIS OF THE CURRENT APPROACH

3.1 Existing System

Currently, users may manage personal expenses using:

- Paper notebooks.
- Manual calculations.
- Spreadsheets.
- Basic calculator applications.
- Separate banking or payment applications.

In a manual approach, the user records income and expenses separately and calculates the remaining balance. Budget monitoring is usually performed manually.

3.2 Strengths of Existing Methods

1. Easy to start using.
2. No specialized software may be required.
3. Users have direct control over their records.
4. Suitable for users with very few transactions.

3.3 Limitations

1. Manual data entry can cause errors.
2. Calculations have to be performed manually.
3. Budget limits are not automatically monitored.
4. Users may forget to record transactions.
5. Generating reports is time-consuming.
6. Spending patterns are difficult to identify.
7. Data stored in notebooks or spreadsheets may not be sufficiently secure.
8. Information may be lost if the physical record or file is damaged.
9. There are generally no automatic budget alerts.

3.4 Identified Gaps

Existing Gap	**Proposed Improvement**
Manual expense tracking	Digital expense recording
Manual categorization	Automatic/assisted categories
Manual budget calculation	Automatic budget monitoring
No automatic alerts	Budget limit notifications
Difficult report generation	Automated financial reports
Scattered information	Centralized database
Security limitations	Authentication and access control
Difficult spending analysis	Category-wise and period-wise summaries

Therefore, a dedicated software system can overcome the major limitations of the current approach by automating financial tracking and analysis.

4. PROPOSED SOFTWARE SOLUTION

4.1 Proposed System

The proposed system is a centralized Personal Expense and Budget Management System that stores financial information in a structured database.

The user can log in, enter income and expenses, categorize transactions, create budgets and monitor financial performance. The system automatically calculates spending and compares it with the defined budget.

4.2 Major Functional Modules

Module 1 – User Management
- User registration.
- Login and logout.
- Profile management.
- Authentication and authorization.

Module 2 – Income Management
- Add income.
- Edit income.
- Delete income.
- View income history.
- Calculate total income.

Module 3 – Expense Management
- Add expenses.
- Edit expenses.
- Delete expenses.
- Categorize expenses.
- View transaction history.

Module 4 – Budget Management
- Create budget.
- Set category-wise limits.
- Set monthly limits.
- Modify budget.
- Monitor budget utilization.

Module 5 – Alert Management
- Detect approaching budget limits.
- Notify users about excessive spending.
- Generate overspending alerts.

Module 6 – Financial Reports
- Monthly expense report.
- Category-wise expense report.
- Income versus expense summary.
- Budget versus actual spending.
- Spending trend analysis.

4.3 Functional Requirements

The system shall:

1. Allow users to register and log in.
2. Allow users to add income.
3. Allow users to add, edit and delete expenses.
4. Allow users to categorize expenses.
5. Allow users to create budgets.
6. Calculate total income and expenses.
7. Calculate remaining budget.
8. Compare actual expenses with budget limits.
9. Generate budget alerts.
10. Generate financial reports.
11. Allow users to view their transaction history.
12. Protect user financial information.

4.4 Non-Functional Requirements

Requirement	Description
Usability	Interface should be simple and easy to understand.
Security	User financial data must be protected.
Reliability	System should provide accurate calculations and consistent results.
Performance	Reports and transaction searches should be generated quickly.
Scalability	System should support increasing numbers of users and transactions.
Maintainability	Modules should be easy to modify and update.
Availability	System should be accessible whenever users need it.

4.5 Application of Software Engineering Principles

Modularity

The system is divided into independent modules such as User Management, Expense Management, Budget Management and Reporting. This makes development and testing easier. **Scalability**

The architecture can support increasing users, transactions and financial records without major redesign. **Maintainability**

Separate modules and well-structured code make it easier to fix errors and introduce new features. **Reliability**

Input validation and proper calculations ensure that income, expenses and budgets are processed accurately.

Usability

A simple dashboard and clear reports help users manage their finances without technical knowledge. **Security**

Authentication, authorization and secure data storage protect sensitive financial information. **Reusability**

Common functions such as authentication, validation and report generation can be reused across different modules.

5. METHODOLOGY / FRAMEWORK JUSTIFICATION

Selected Methodology: AGILE – SCRUM

The Agile Scrum methodology is suitable for developing the Personal Expense and Budget Management System because the requirements may change based on user feedback.

Why Agile Scrum is Suitable

1. Incremental Development

The system can be developed in small increments.

For example:

- Sprint 1 – User registration and login.
- Sprint 2 – Income and expense

management.
- Sprint 3 – Budget management. **2. Continuous User Feedback**
- Sprint 4 – Alerts.
- Sprint 5 – Reports and dashboard.
- Sprint 6 – Testing and improvemen

Users can test each completed module and provide feedback. The development team can incorporate the feedback into subsequent sprints.

3. Early Delivery

Basic functionality can be delivered early instead of waiting until the entire application is completed. **4. Easy Requirement Changes**

If users request features such as recurring expenses, additional categories or improved reports, they can be added to the product backlog and implemented in later sprints.

5. Continuous Testing

Each sprint includes development and testing, helping identify defects at an early stage. **6. Better Risk Management**

Problems can be identified during individual sprints instead of only at the end of the project.

Scrum Roles

Role	Responsibility
Product Owner	Defines requirements and prioritizes features.
Scrum Master	Facilitates Scrum activities and removes development obstacles.
Development Team	Designs, develops and tests the software.

Scrum Activities

- Product Backlog
- Sprint Planning
- Daily Scrum
- Sprint Development
- Sprint Review
- Sprint Retrospective

Agile Scrum therefore provides an iterative, flexible and user-focused approach for developing and maintaining the proposed system.

6. EXPECTED OUTCOMES AND LIMITATIONS

6.1 Expected Outcomes

The proposed system is expected to:

1. Reduce manual effort in expense management.
2. Improve the accuracy of financial calculations.
3. Help users maintain organized financial records.
4. Improve budget monitoring.

6.2 How the Solution Addresses the Challenges

5. Provide timely overspending alerts. 6. Help users understand their spending patterns.
7. Generate useful financial reports. 8. Improve financial decision-making. 9. Provide secure access to personal financial information.

Challenge	System Solution
Forgetting transactions	Easy digital transaction recording
Calculation errors	Automatic calculations
Overspending	Budget monitoring and alerts
Poor spending visibility	Reports and summaries
Scattered records	Centralized database
Security concerns	Authentication and access control
Difficult analysis	Category-wise and time-based reports

6.3 Potential Risks

1. Incorrect data entered by users.
2. Unauthorized access to financial information.
3. Database failure or data loss.
4. Software bugs.
5. Users may not regularly update transactions.
6. Performance issues when the number of transactions increases.
7. User resistance to adopting a new system.

6.4 Implementation Challenges

- Designing a secure database.
- Protecting sensitive financial information.
- Maintaining accurate calculations.
- Creating a simple user interface.
- Implementing reliable notification mechanisms.
- Testing the system with different financial scenarios.

6.5 Limitations

1. The accuracy of reports depends on the accuracy of user-entered data.
2. The system may not automatically know cash transactions unless the user records them.
3. Advanced financial prediction may require additional data and algorithms. 4. External banking integration may require third-party services.
5. Security threats cannot be completely eliminated, although appropriate security controls can

reduce the risk.

6.6 Future Enhancements

Future versions can include:

- Automatic bank transaction integration.
- AI-based spending prediction.
- Personalized saving recommendations.
- Recurring expense management.
- Investment tracking.
- Multi-currency support.
- Mobile application support.
- Advanced financial dashboards.
- Voice-based expense entry.
- Intelligent anomaly detection for unusual transactions.

CONCLUSION

The Personal Expense and Budget Management System provides a systematic software-based solution for managing income, expenses and budgets. The proposed system addresses the major limitations of manual financial management through automated calculations, categorization, budget monitoring, alerts and financial reporting.The system follows important software engineering principles such as modularity, scalability, maintainability, reliability, usability and security. Agile Scrum is selected because it supports incremental development, continuous testing, changing requirements and user feedback. Therefore, the proposed solution is feasible, maintainable, scalable and user-oriented.